

AN EXCEL BASED HEART DISEASE ANALYSIS

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1. INTRODUCTION

Heart disease is one of the major health concerns, and analysing patient data can help identify important patterns and risk factors. This project, “Heart Disease Analysis using Excel,” focuses on analysing a dataset containing information about patients. The dataset includes variables such as age, gender, chest pain type, resting blood pressure, cholesterol level, maximum heart rate, ST depression, exercise-induced angina, and heart disease status.

The main objective of this project is to clean and organize the raw data and convert it into meaningful information. Data cleaning techniques are used to handle missing values, inconsistent formats, duplicate records, and incorrect entries. After cleaning, the data is analysed using Excel tools such as sorting, filtering, Pivot Tables, and formulas.

Different types of charts, including pie charts, bar charts, column charts, stacked charts, and scatter charts, are used to present the results visually. These visualizations make it easier to compare heart disease cases based on gender, age group, chest pain type, cholesterol, blood pressure, maximum heart rate, and other factors.

Overall, this project demonstrates how Microsoft Excel can be used as a Business/Data Analytics tool to transform raw patient data into clear and useful insights.

2. PROBLEM STATEMENT

Heart disease is a major health concern, and patient data contains various factors that may be associated with heart disease. However, raw patient data may contain missing values, inconsistent formats, and incorrect entries, making it difficult to analyse directly. Therefore, this study aims to clean, organize, and analyse patient data using Microsoft Excel to identify meaningful patterns and relationships between heart disease and factors such as age, gender, chest pain type, blood pressure, cholesterol, maximum heart rate, ST depression, and exercise-induced angina.

2.1 Objectives of the Study

1. The main objectives of the study are:
2. To clean and organize the raw heart disease dataset.
3. To identify and handle missing and inconsistent data.
4. To analyse the relationship between heart disease and different patient characteristics.
5. To study factors such as age, gender, chest pain, blood pressure, cholesterol, and maximum heart rate.
6. To use Excel tools such as Sorting, Filtering, Pivot Tables, and formulas for data analysis.
7. To present the findings using different charts and visualizations.
8. To generate meaningful insights from patient data using Microsoft Excel.

2.2 Scope of the Study

1. To analyse heart disease patient data.
2. To study age and gender factors.
3. To analyse blood pressure and cholesterol.
4. To study chest pain types.
5. To analyse maximum heart rate.
6. To study exercise-induced angina.
7. To identify patterns related to heart disease.
8. To present findings using Excel charts.

3.DATASET DESCRIPTION

3.1 Data Source

The dataset used for this study was obtained from Kaggle. The dataset contains patient information related to heart disease. According to the project file, the dataset consists of 1,021 rows and 10 columns.

3.2 OVERVIEW

- Dataset: Heart Disease Dataset
- Source: Kaggle
- Total Records: 1,021 patients
- Variables: Patient-related data
- Main Factors: Age, Gender, Chest
- Pain, Blood Pressure, Cholesterol,
- Max Heart Rate, ST Depression, Angina
- Outcome: Heart Disease Status

Patient_ID	Age	Gender	Chest_Pain_Type	Resting_BP_mmHg	Cholesterol_mg/dl	Max_Heart_Rate	ST_Depression	Exercise_Induced_Angina	Heart_Disease
524	87			151.3430808	162.3782381	162.2876412	2.801430099	Yes	zero
603	66	male		101.2140983	218.380992	155.8462006	0.182317603	No	zero
527	87	FEMALE	Atypical Angina	150.3540933	248.7656229	195.3351925	0.875113115	yes	1
32	87	FEMALE	Non-Anginal Pain	124.8653892		180.7303696	0.30050478	yes	
617	45		Typical Angina	112.4373025	211.5722961	121.9689065	N/A	No	one
586	77	Male		105.6909087	137.2538666	141.3174428	2.202576505	Yes	
445	89	FEMALE	Asymptomatic	130.0572093	252.0862332	164.8872724	1.889153825	No	one
733	59	male	Atypical Angina	120.4919549		134.8131928	1.842297327	Yes	
77	43	Male	Typical Angina	125.5093097	162.8757884	120.0075618	2.592669998	No	
412	75	Male	Non-Anginal Pain	118.4516737	173.5333944	157.9630251	-0.341064713	No	zero
569	84	Male	Asymptomatic	114.6248977	147.492963	90.20109489	2.784775039		one
71	89	FEMALE	Non-Anginal Pain	131.0821572	147.0546206	109.0492736	1.026204753	yes	
676	41	FEMALE		114.1935045	170.5364174	172.5133269	1.21780234	yes	one
102	52		Non-Anginal Pain	92.54050655	192.4581144	138.7943846	0.593114755		
895	61	FEMALE	Typical Angina	99.33573749		147.488107	3.035873535	yes	0
926	77	FEMALE	Typical Angina	134.9528816	168.0858637	195.5135977	1.193379452		1
630	83	Female	Atypical Angina		201.501574	112.2029145	1.659054344	No	one
300	45		Atypical Angina	138.9288225	238.8703676	129.1429547	2.253132825	Yes	0
925	57	Female		141.9626586	158.4022541	120.5759472	0.331856952	yes	zero
425	62	Female	Asymptomatic	129.2365672	238.7577747	140.9220835	1.041287456		one
97	60			122.8437242	219.3161659	185.7214427	0.131739038	Yes	
	57			148.0644776	217.3354989	171.8206088	0.827370866	NO	one
261	66	Male	Asymptomatic	129.5707665	163.4521862	175.2367947	1.827115429	yes	1

Column Name	Description	Data Type
Patient ID	Unique identification number assigned to each patient.	Integer
Age Category	Category representing the patient's age group.	String
Age	Age of the patient in years.	Integer
Gender	Gender of the patient (Male, Female, or Unknown).	String
Chest Pain Type	Type of chest pain experienced by the patient.	String
Resting BP mmHg	Resting blood pressure measured in millimeters of mercury (mmHg).	Float
Cholesterol mg/dl	Cholesterol level measured in milligrams per deciliter (mg/dL).	Float
Max Heart Rate	Maximum heart rate achieved by the patient.	Integer
ST Depression	ST depression value observed during exercise testing.	Float
Exercise Induced Angina	Indicates whether the patient experienced angina during exercise (Yes/No).	String
Heart Disease	Indicates the presence or absence of heart disease (typically 0 = No, 1 = Yes).	Integer

Dataset Source: Kaggle

Total Row: 1021

Total Column:10

3. DATA CLEANING

3.1 IMPORTING THE DATASET

- Loaded the dataset into excel
- All columns were reviewed to understand their structure, Data types, and relevance to the analysis

3.2.Alt +H+O+I: AutoFit Column width

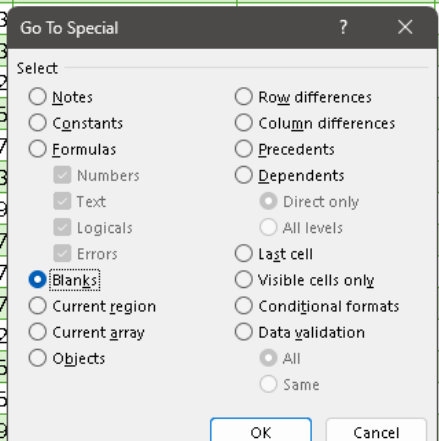
3.3 Alt +H+O+A: AutoFit Row height

3.4.checking for missing values: The Data Set checked for blank or missing values. Filters were used to identify empty cells in different columns. Patient ID, Age, Gender, Chest Pain Type, Resting Bp mmHg, Cholesterol mg/di, Max Heart Rate, Exercise Induced Angina, Heart Disease.

3.5. Handling Missing Value

- **Patient id** — Go To Special, select blank, Blank fill with “0”

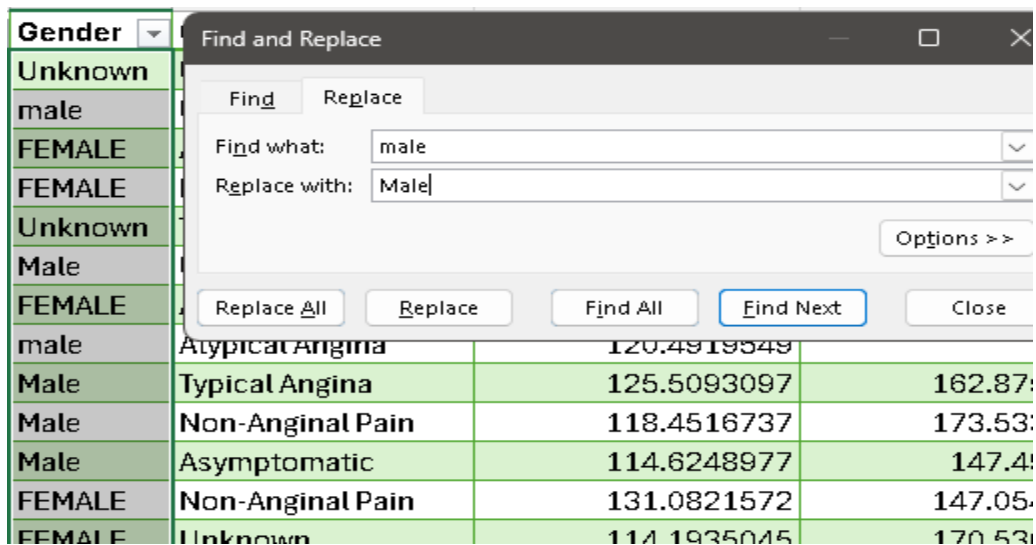
Patient_ID	Age	Gender	Chest_Pain_Type	Resting_BP_mmHg	Cholesterol_mg/dl	Max_Heart_Rate
524	87			151.3430808	162.3782381	162.2876412
603	66	male		101.2140983		
527	87	FEMALE	Atypical Angina	150.3540933		
32	87	FEMALE	Non-Anginal Pain	124.8653892		
617	45		Typical Angina	112.4373025		
586	77	Male		105.6909087		
445	89	FEMALE	Asymptomatic	130.0572093		
733	59	male	Atypical Angina	120.4919549		
77	43	Male	Typical Angina	125.5093097		
412	75	Male	Non-Anginal Pain	118.4516737		
569	84	Male	Asymptomatic	114.6248977		
71	89	FEMALE	Non-Anginal Pain	131.0821572		
676	41	FEMALE		114.1935045		
102	52		Non-Anginal Pain	92.54050655		
895	61	FEMALE	Typical Angina	99.33573749		
926	77	FEMALE	Typical Angina	134.9528816	166.0858637	195.5135977
630	83	Female	Atypical Angina		201.501574	112.2029145
300	45		Atypical Angina	138.9288225	238.8703676	129.1429547



Patient_ID				
838				
245				
287				
290				
599				
762				
60				
623	65	Female	Asymptomatic	103.6913109
319	36	Male	Atypical Angina	216.6889267
11	52	FEMALE		88.17217651
		Female	Asymptomatic	119.7832159
332	79	Male	Typical Angina	132.9359021
706	86	male	Asymptomatic	122.150827
87	33	Male	Asymptomatic	134.2546139
804	87	FEMALE	Non-Anginal Pain	142.3086925

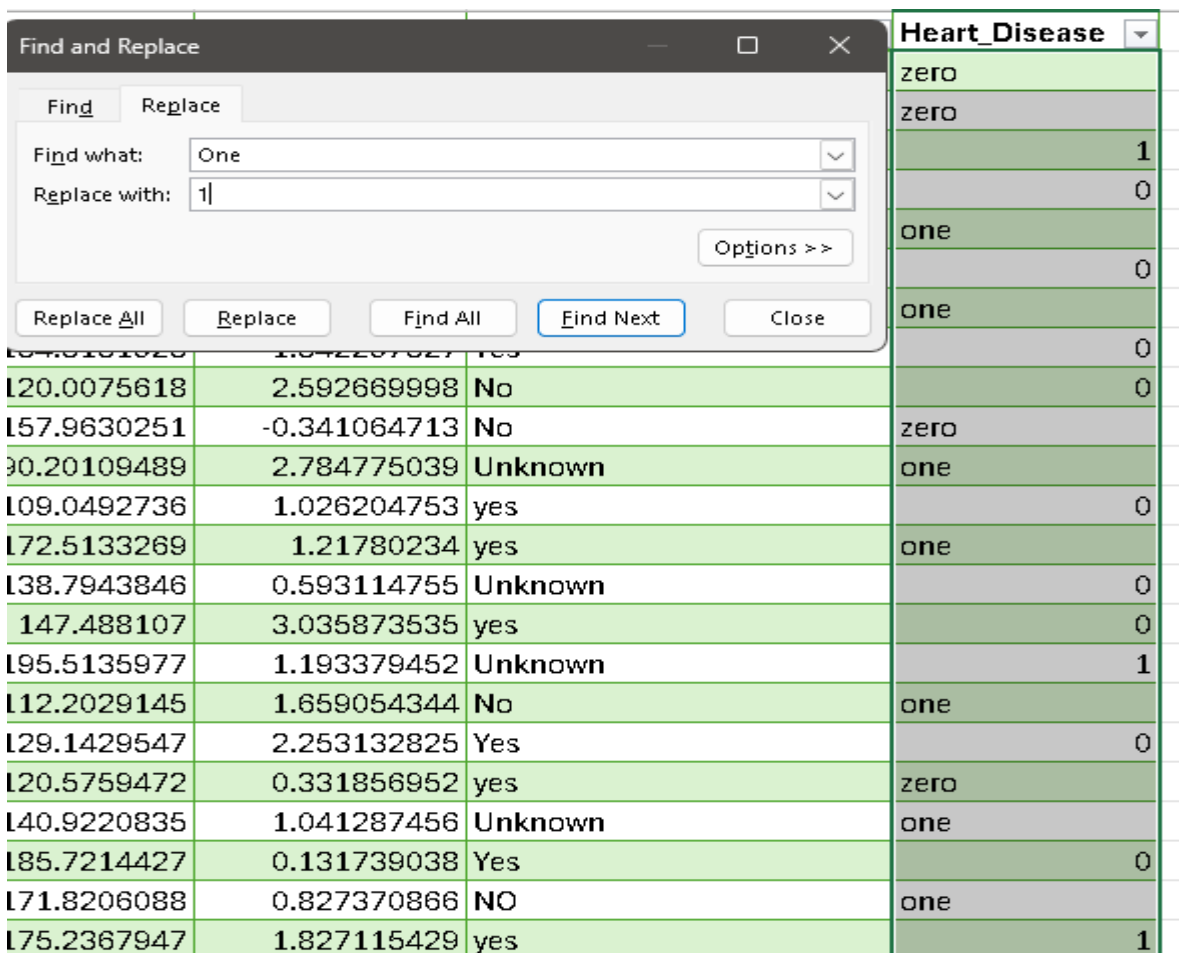
- **Gender** -Categorical values such as “FEMALE” and “male” were standardized by replacing them with the abbreviations “Female” and “Male”. This made the dataset more consistent and easier to analyze.

Gender				
Unknown				
Male				
FEMale				
FEMale				
Unknown				
Male				
FEMale				
Male	Atypical Angina		120.4919549	
Male	Typical Angina		125.5093097	162.87578
Male	Non-Anginal Pain		118.4516737	173.53339
Male	Asymptomatic		114.6248977	147.4929
FEMale	Non-Anginal Pain		131.0821572	147.05462
FEMale	Unknown		114.1935045	170.53642
Unknown	Non-Anginal Pain		92.54050655	192.45812
FEMale	Typical Angina		99.33573749	



Press Ctrl + H → Enter Female in “Find what” → Enter F in “Replace with” → Click “Replace All”.

- **Heart disease** - Categorical values such as “One” and “zero” were standardized by replacing them with the abbreviations “1” and “0”. This made the dataset more consistent and easier to analyze.



Find and Replace			Heart_Disease
Find what: One			zero
Replace with: 1			zero
Options >>			1
Replace All			0
Replace			one
Find All			0
Find Next			one
Close			0
120.0075618	2.592669998	No	0
157.9630251	-0.341064713	No	zero
90.20109489	2.784775039	Unknown	one
109.0492736	1.026204753	yes	0
172.5133269	1.21780234	yes	one
138.7943846	0.593114755	Unknown	0
147.488107	3.035873535	yes	0
195.5135977	1.193379452	Unknown	1
112.2029145	1.659054344	No	one
129.1429547	2.253132825	Yes	0
120.5759472	0.331856952	yes	zero
140.9220835	1.041287456	Unknown	one
185.7214427	0.131739038	Yes	0
171.8206088	0.827370866	NO	one
175.2367947	1.827115429	yes	1

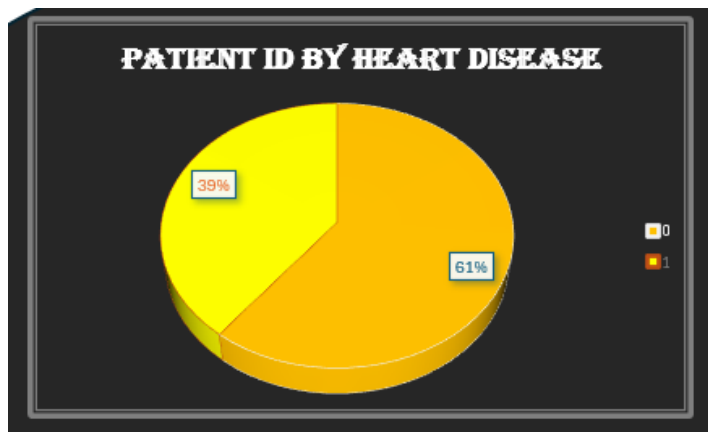
- Gender - Blank fill with “Unknown”

D	E	F	G
Gender			
Male	Unknown		
Female	Unknown		
Female	Unknown		
Male	Unknown		
Female	Unknown		
Male	Unknown		
Male	Typical Angina	118.4516737	173.5333944
Male	Non-Anginal Pain	114.6248977	147.492963
Male	Asymptomatic	131.0821572	147.0546206
Female	Non-Anginal Pain	114.1935045	170.5364174
Female	Unknown	92.54050655	192.4581144
Female	Non-Anginal Pain	99.33573749	0
Female	Typical Angina	134.9528816	168.0858637
Female	Atypical Angina	0	201.501574
Female	Atypical Angina	138.9288225	238.8703676
Female	Unknown	141.9626586	158.4022541
Female	Asymptomatic	129.2365672	238.7577747
Unknown	Unknown	122.8437242	219.3161659
Unknown	Unknown	148.0644776	217.3354989
Male	Asymptomatic	129.5707665	163.4521862
Male	Asymptomatic	108.1137012	179.3197233
Male	Unknown	117.1494145	248.9211056
Female	Atypical Angina	112.8174382	185.7807675
Female	Asymptomatic	105.9014535	130.1440907

4. PIVO TABLE AND CHART

1. What is the distribution of patients based on heart disease status?

Row Labels	Count of Patient_ID
0	618
1	402
Grand Total	1020

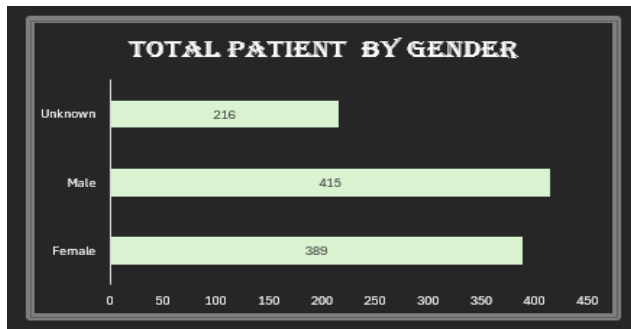


Interpretation:

The chart shows that 62% of patients do not have heart disease, while 38% of patients have heart disease. Therefore, the majority of patients in the dataset are without heart disease.

2. How are the patients distributed according to gender?

Row Labels	Count of Patient_ID
Female	389
Male	415
Unknown	216
Grand Total	1020



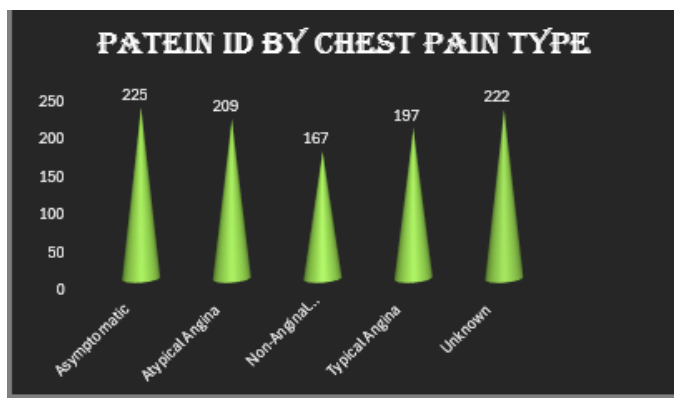
Interpretation:

The chart shows the distribution of patients among Male, Female, and Unknown categories. Male patients represent the largest group, followed by female patients, while a smaller number of records have an unknown gender.

3. How are patients distributed according to chest pain type?

Row Labels	Count of Patient_ID
Asymptomatic	225
Atypical Angina	209
Non-Anginal Pain	167
Typical Angina	197
Unknown	222
Grand Total	1020

4.

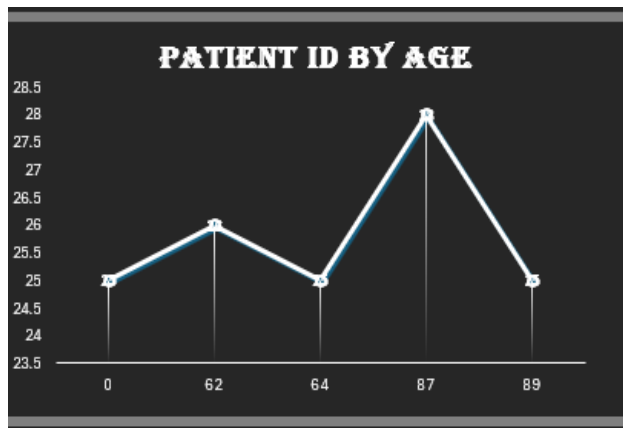


Interpretation:

The chart shows that patients are distributed across different chest pain types, including Asymptomatic, Atypical Angina, Non-Anginal Pain, Typical Angina, and Unknown. The distribution is relatively balanced, with no single chest pain category showing an extremely large difference from the others.

4. How does the number of patients vary across different age categories?

Row Labels	Count of Patient_ID
0	25
62	26
64	25
87	28
89	25
Grand Total	129

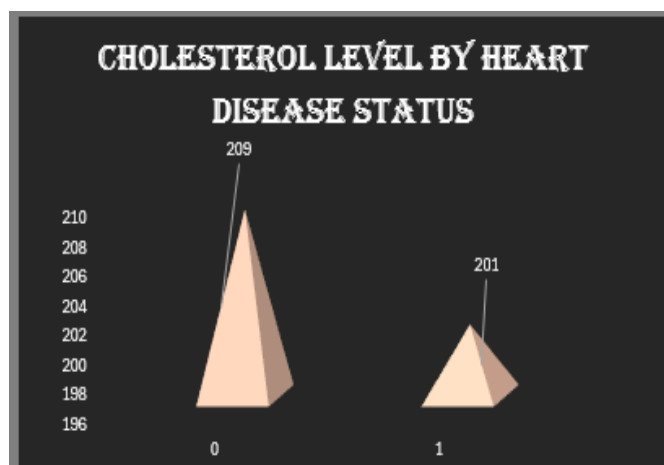


Interpretation:

The chart shows variation in the patient distribution across the different age categories. Some age categories have slightly higher values than others, indicating that the patients are not equally distributed across all age groups.

5. How does cholesterol level vary according to heart disease status?

Row Labels	Average of Cholesterol_mg/dl
0	208.583349
1	200.8135962
Grand Total	205.5211523

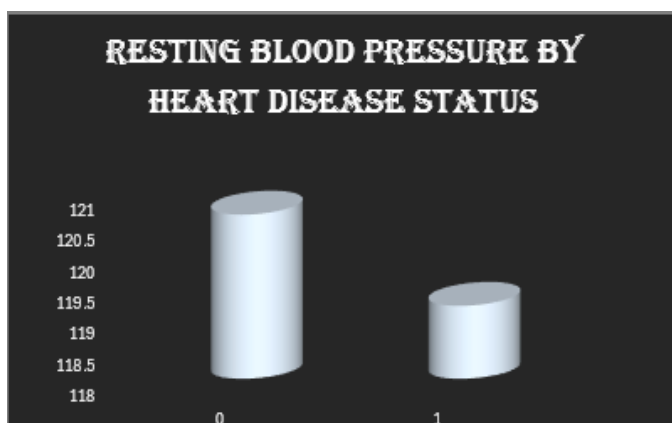


Interpretation:

The chart compares cholesterol levels between patients with and without heart disease. The cholesterol levels appear to be relatively close between the two groups, indicating that there is no major difference in the displayed average cholesterol level based on heart disease status.

6. How does resting blood pressure differ between patients with and without heart disease?

Row Labels	Average of Resting_BP_mmHg
0	120.6303137
1	119.1690488
Grand Total	120.0544034

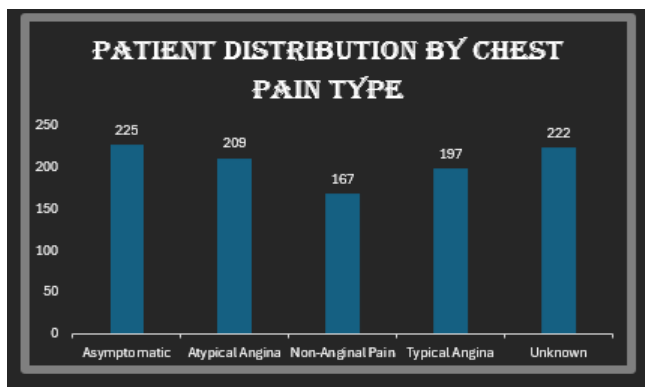


Interpretation:

The chart compares resting blood pressure for the two heart disease groups. The values are relatively close, although the group without heart disease shows a slightly higher resting blood pressure in the displayed analysis.

7.How is maximum heart rate distributed according to heart disease status?

Row Labels	Average of Max_Heart_Rate
0	149.1030431
1	145.1334648
Grand Total	147.5385622

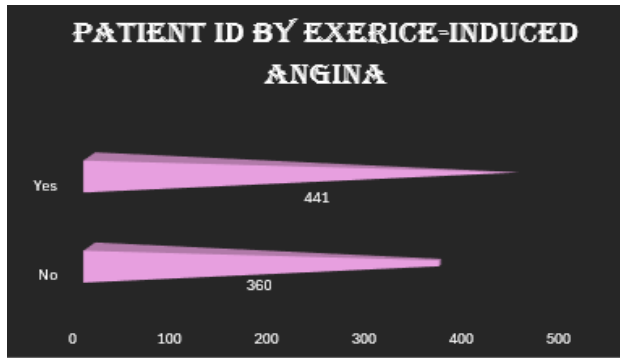


Interpretation:

The chart shows that maximum heart rate is distributed between patients with and without heart disease. The two categories have almost similar proportions, with the 0 (No heart disease) category being slightly higher than the 1 (Heart disease) category.

8.How many patients experienced exercise-induced angina?

Row Labels	Count of Patient_ID
No	360
UnkNown	219
Yes	441
Grand Total	1020

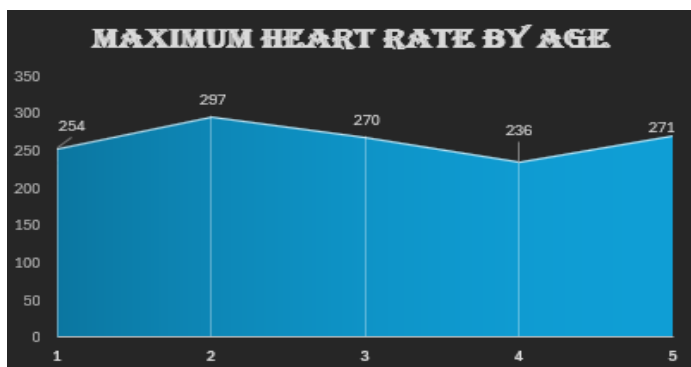


Interpretation:

The chart shows the number of patients with and without exercise-induced angina. Patients without exercise-induced angina form a slightly larger group, while a considerable number of patients experienced angina during exercise.

9. How are patients distributed across different heart disease status categories?

Row Labels	Max of Max_Heart_Rate
32	253.7341116
64	297.1292173
67	269.5723809
87	235.8273001
89	270.6823017
Grand Total	297.1292173

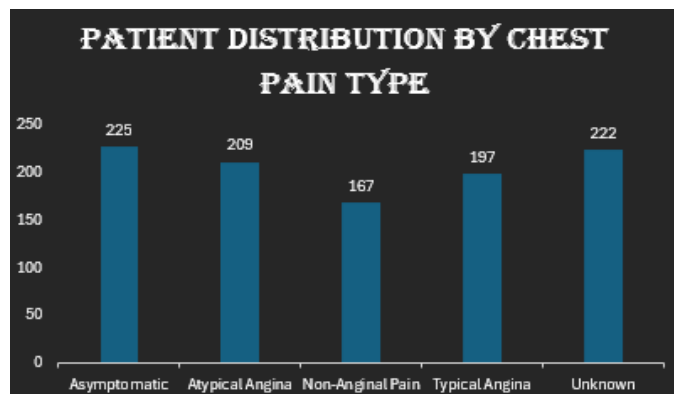


Interpretation:

The chart compares the number of patients according to heart disease status. It shows that the No heart disease (0) category has a higher number of patients than the Heart disease (1) category.

10. What is the distribution of patients across different chest pain types?

Row Labels	Count of Patient_ID
Asymptomatic	225
Atypical Angina	209
Non-Anginal Pain	167
Typical Angina	197
Unknown	222
Grand Total	1020

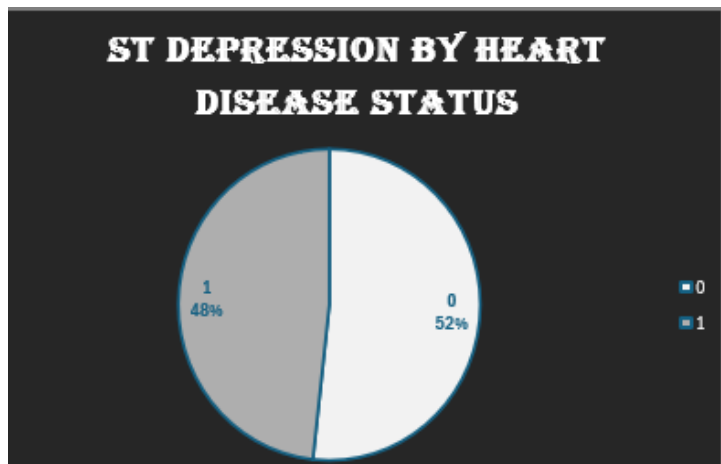


Interpretation:

The chart shows the number of patients for each chest pain category. Asymptomatic and Unknown categories have relatively higher patient counts, while Non-Anginal Pain has a comparatively lower count. This indicates variation in chest pain types among the patients.

11. How is ST depression distributed according to heart disease status?

Row Labels	Average of ST_Depression
0	0.990831181
1	0.924083519
Grand Total	0.96452475

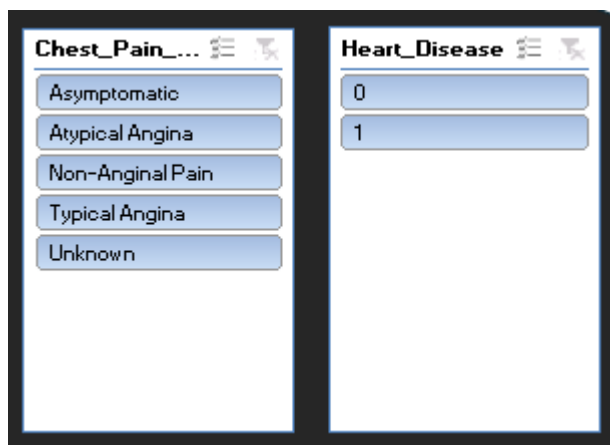


Interpretation:

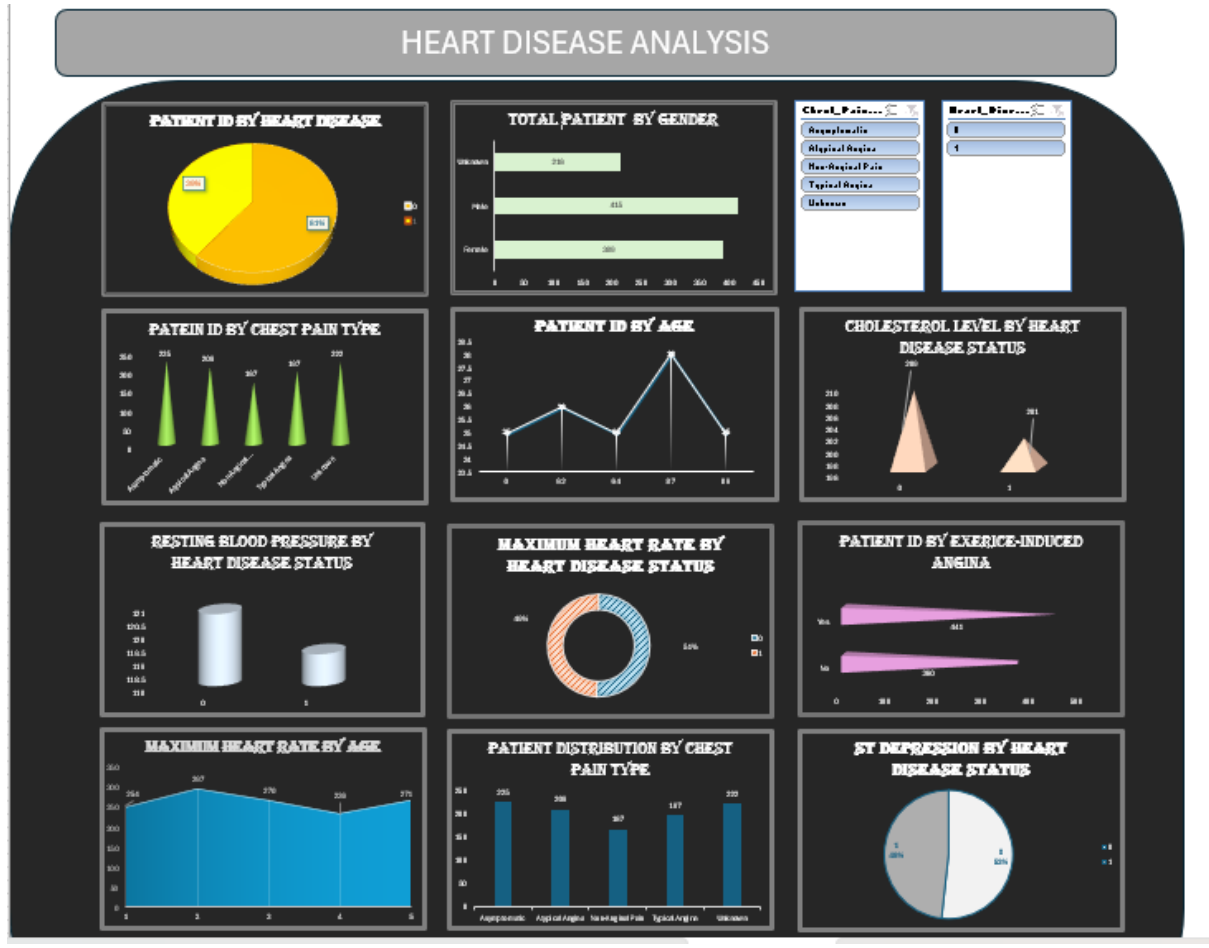
The chart shows that approximately 62% of the observations belong to heart disease status 0, while around 38% belong to status 1. Thus, the majority of patients in the dataset fall under the no-heart-disease category.

SLICER

A Slicer is an interactive filtering tool in Excel that allows users to quickly filter and analyze data based on selected categories



5.DASHBOARD



5. CONCLUSION

The study successfully analysed heart disease patient data using Microsoft Excel. The data was cleaned and organized before analysis. Various Excel tools such as Pivot Tables, charts, and slicers were used to identify and present patterns in the data. The dashboard provides a simple and interactive way to understand patient information based on factors such as age, gender, chest pain, blood pressure, cholesterol, and heart disease status. Overall, the project demonstrates the usefulness of Excel for data cleaning, analysis, and visualization.